SEC P vs NP log 2026-04-29

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 08:50 UTC

SEC P vs NP — daily log 2026-04-29

17 cycles recorded

GF(2)-Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid theory — the GF(2) vector matroid M_F realized by the term-indicator matrix A_F ∈ F_2^{m×N} of a monotone DNF F = ∨*{i=1}^m T_i (rows* ▯{T_i}, columns indexed by variables). Its rank ρ_2(F) := rank_{F_2}(A_F) is the canonical computable matroid invariant in the Whitney–Welsh tradition; the corresponding ‘rank defect’ Δ(F) := log_2(m(F)+1) − log_2(ρ_2(F)+1) is a polymatroid-style quantity (subadditive log-rank vs. log-cardinality) that is rarely deployed in monotone circuit lower bounds (Razborov 1985 used Bollobás set-pair / sunflower closure, not GF(2) span of clutter indicators). × Monotone DNF representation size of the k-CLIQUE_n indicator on N = C(n,2) edge variables; equivalently, the minimum-cardinality clutter of minimal terms whose union gives k-CLIQUE_n.
- **Recorded**: 2026-04-29 00:11 UTC
- **Entry ID**: `023dd98c00ee`

Statement

Let F be a monotone DNF on N variables with minimal-term family H_F = {T_1,…,T_m}, A_F its m×N indicator matrix, and define Δ(F) := log_2(m+1) − log_2(rank_{F_2}(A_F)+1). Conjecture: (a) Δ(F∧G) ≤ Δ(F) + Δ(G) + 1 and Δ(F∨G) ≤ max(Δ(F),Δ(G)) + 1 (approximate submodularity through Boolean operations); (b) Δ(F) ≤ log_2(N+1) for any monotone DNF with m ≤ poly(N) (since rank_{F_2} ≥ 1 and m ≤ poly); (c) for the canonical k-CLIQUE_n DNF F^{n,k} with k = ⌊n/2⌋, Δ(F^{n,k}) ≥ n/2 − 3·log_2 n − 4 for all n ≥ 6, witnessing a Ω(n) gap between log-cardinality and log-GF(2)-rank.

Rationale

The GF(2) span of clique-edge-indicator vectors collapses to a sub-quadratic-rank object (rank ≤ C(n,2)), while the minimal-term count C(n,⌊n/2⌋) ≈ 2^n/√n is exponential — so the LOG ratio is Θ(n), much larger than any rank deficit attainable by poly-size clutters. Razborov’s approximation method tracks subadditive measures across AND/OR gates; an honest submodular Δ that is forced to Ω(n) on the clique clutter is exactly the missing ingredient for a Razborov-style argument that does not require sunflower closures, since GF(2)-rank is robust under union operations on supports. The bridge from matroid representability to monotone DNF size is concrete (one Gaussian elimination per gate), avoiding the natural-proofs / algebrization barriers that block spectral or Fourier approaches.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2210.12983v2] Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 118, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 78, in run_trial
   F_random = random_poly_dnf(n, n, 2) + random_poly_dnf(n, n**2, 3) + random_poly_dnf(n, n * math.floo
                ^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 56, in random_poly_d
    selected_edges = random.sample(edge_set, w)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^

   File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError in random_poly_dnf (random.sample called on a non-sequence) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix random_poly_dnf by passing a list (e.g., list(edge_set) or sorted(edge_set)) to random.sample and rerun the 30-seed sweep.

Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity Across Bit-Permuted Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe Algebras (within Geometric Group Theory) × Deterministic 2-party communication complexity, specifically the multiplicity m_Π of the RoeCover extracted from a deterministic protocol's ProtocolPullback for the lifted function f∘G^n
- **Recorded**: 2026-04-29 00:24 UTC
- **Entry ID**: `aa9fe4edbfe5`

Statement

Let G_1 = (X, Y, g, d) be a MetricGadget on small alphabet (|X|,|Y| ≤ 4) and let G_2 be obtained from G_1 by a bi-Lipschitz bijection φ: X×Y → X×Y with distortion ≤ 2 (a coarse equivalence: φ permutes inputs while distorting d by at most a factor 2). Fix any boolean f: {0,1}^n → {0,1} with n ≤ 4. Then for every deterministic protocol Π_1 computing f∘G_1^n, there exists a deterministic protocol Π_2 computing f∘G_2^n with cost cost(Π_2) ≤ cost(Π_1) + O(n), and the multiplicities of their CoarsePullback covers at scale R = diam(G) satisfy m_{Π_2} ≤ 2 · m_{Π_1} + 1. Equivalently, CLC(f,G_1) and CLC(f,G_2) agree up to a constant factor independent of f, instantiating Axiom A1

(Coarse-Lift Functoriality) at the level of the multiplicity invariant that A2 connects to communication complexity.

Rationale

This conjecture directly tests Axiom A1 (Coarse-Lift Functoriality) by asserting that the *multiplicity invariant* m_Π exposed by Axiom A2 (Protocol → Cover) is preserved under coarse equivalences of the underlying gadget. If A1 holds, a bi-Lipschitz φ on $X \times Y$ lifts coordinatewise to a bi-Lipschitz map on $(X \times Y)^n$ with the $d_\oplus$ metric (distortion preserved since ℓ^1 products of bi-Lipschitz maps remain bi-Lipschitz), so any Roe-style cover at scale R for the G_1-lift can be transported to a cover at scale $\leq 2R$ for the G_2-lift while at most doubling the multiplicity (each transported set may merge with one neighbor at the boundary, contributing the +1 slack). The O(n) cost overhead in Π_2 reflects simulating φ coordinatewise. Verifying this empirically would falsify A1 if multiplicities diverge, and would strengthen confidence that CLC is genuinely a coarse invariant — a prerequisite for using it as a lower-bound technique against communication complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [1706.00276v1] Counting coarse subsets of a countable group - [2110.02008v2] Lifted Reed-Solomon Codes and Lifted Multiplicity Codes - [1903.06084v8] Lifting coarse homotopies

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 49, in run_trial
    phi_perms = generate_bi_lipschitz_permutations()
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 35, in generate_bi_l
    if is_bi_lipschitz(phi):
       ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 24, in is_bi_lipschi
    if dist(phi[(x1, y1)], phi[(x2, y2)]) > distortion * dist((x1, y1), (x2, y2)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 18, in dist
    return max(abs(ai - bi) for ai, bi in zip(a, b))
                                          ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in the bi-Lipschitz check (dist() called on integers instead of tuples) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the dist() helper to handle the (x,y) pair representation correctly (ensure phi maps tuples to tuples and dist unpacks both arguments), then rerun the 5-seed sweep.

Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real Rank

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent homology / Vietoris-Rips topology of finite metric spaces (Vietoris 1927; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson-Zomorodian 2005; Carlsson 2009). The 0-th persistent Betti number β_0(VR(X,ε)) of a finite metric space (X,d) at scale ε is the number of connected components of the ε-thresholded distance graph, computable by Kruskal/union-find in O(|X|^2 α(|X|)). Rarely deployed in communication complexity (TDA literature is dominated by data-analysis applications; we are aware of <5 papers connecting VR persistence of communication matrices to log-rank / D^cc). × Real rank rk_R(M_f) of the 2^k × 2^k ±1 communication matrix M_f[x,y]=(-1)^{f(x,y)} of a Boolean function f:{0,1}^k × {0,1}^k → {0,1} — the canonical Lovász log-rank / Mehlhorn-Schmidt target for deterministic two-party communication complexity D^cc(f) ≥ log_2 rk_R(M_f).
- **Recorded**: 2026-04-29 00:40 UTC
- **Entry ID**: `bc750201aaf2`

Statement

Let X_f^± ⊂ $\{\pm1\}^{\{2}$k}/{±1} be the set of distinct rows of M_f modulo global sign, equipped with the projective Hamming distance d_±(r,s) = min(d_H(r,s), 2^k - d_H(r,s)). Let τ_±(M_f) := β_0(VR(X_f^±, 2^{k-2})), the number of components of the graph linking r,s whenever d_±(r,s) ≤ 2^{k-2}. Then for every Boolean f and every k ∈ {3,4,5,6}, τ_±(M_f) ≤ rk_R(M_f); a single (k,f) with τ_±(M_f) > rk_R(M_f) refutes.

Rationale

Rank-r ±1 matrices have all rows lying in an r-dimensional subspace of R^N, and ±1 vectors in such a subspace are constrained to a structured (centrally-symmetric) finite set whose pairwise distances avoid the 'mid-Hamming' regime around N/2 only by sitting in a small affine pattern; persistence at scale N/4 should therefore not be able to manufacture more than r distinct components after the ±-quotient. This bridges the topological 'shape-counting' of the row metric space to the algebraic rank-bound side of log-rank, a structural inequality complementary to spectral approaches and untouched by natural-proofs (rk_R is the property being lower-bounded, not a hardness predicate on truth tables) and untouched by algebrization (no oracle relativization is used).

Novelty

- Judge: `NOVEL` over 5 arXiv hits

Top hits consulted: - [LIPIcs.ESA.2024.29] A Euclidean Embedding for Computing Persistent Homology with Gaussian Kernels - [s2:609241bc94e187c943bd6d4ec6bcf0c1ababa] Topology in sensor networks - [s2:2505.16879] How high is 'high'? Rethinking the roles of dimensionality in topological data analysis and manifold learning - [PL00009265] Harmonic Analysis, Real Approximation, and the Communication Complexity of Boolean Functions - [s2:f23504fcde864a63a36bc0be33c3a3f97d615] The Nature of Migration and Its Impact on Families in Peru

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 163, in run_trial
    M.append([Fraction(row[i]) for i in range(n)])
```

```
                              ~~~^^^
IndexError: tuple index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition could be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the row-construction bug (tuple index out of range when building M from row entries) and re-run the full 4×30×6 grid to obtain τ_± and rk_R counts.

Secant Rank Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry of secant varieties × Communication complexity
- **Recorded**: 2026-04-29 01:14 UTC
- **Entry ID**: `1a21dd39fb6f`

Statement

The secant rank of the communication matrix for the Disjointness problem DISJ_n is Ω(n), implying a Ω(n) lower bound on randomized communication complexity.

Rationale

Secant rank measures the minimal number of rank-1 tensors needed to approximate a matrix, directly linking to communication complexity via the log-rank conjecture. For DISJ_n, its matrix structure forces high secant rank due to combinatorial constraints, providing a novel algebraic invariant for proving lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.82s`

```
me': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.5, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
RESULT: SUPPORTED mean=8373.511111111111 std=0 support_fraction=1.0
```

Judge reasoning

The test used n=6, which is too small to establish Ω(n) bounds. Metric values show no scaling with n, suggesting potential saturation or hidden structure in small cases. | next: Test with larger n (e.g., n=128) to observe scaling behavior of secant rank

---

Sign-Rank Distinguishing Tensor for Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: discrepancy theory × matrix rigidity
- **Recorded**: 2026-04-29 01:46 UTC
- **Entry ID**: `d2ba6b1a27c2`

Statement

For any read-twice branching program P computing a Boolean function f, the sign-rank of its communication matrix M_P is O(log size(P)), but for the inner product mod 2 function IP_2, the sign-rank of its trivial read-twice BP is Ω(n).

Rationale

This conjecture leverages the connection between sign-rank (a discrepancy-theoretic invariant) and matrix rigidity (a field_B concept). The IP_2 function's exponential lower bound for read-once BPs suggests its read-twice representation must exhibit high sign-rank, while read-twice BPs for general functions should have logarithmic sign-rank due to their structured computation. This bridges the gap between BP complexity and communication complexity barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 77, in run_trial
    A = generate_random_sign_matrix(n, rank)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 47, in generate_rand
    U = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 30, in gaussian_elim
    factor = Fraction(A[j][i], A[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing data collection | next: Debug gaussian_elimination() to handle zero pivots and regenerate matrices

---

SOS Hierarchy Integrality Gap for Random CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares hierarchy × Constraint Satisfaction Problems (CSPs)
- **Recorded**: 2026-04-29 02:17 UTC
- **Entry ID**: `90b9c49aa839`

Statement

For any constant ε > 0, there exists a CSP instance with n variables and m = Θ(n) constraints such that the SOS hierarchy requires Ω(n^{1-ε}) levels to achieve an integrality gap of 1 - ε for its relaxation.

Rationale

This conjecture builds on the SOS hierarchy's known limitations in handling random CSPs, where the hierarchy's ability to capture exact solutions scales sublinearly with the problem size. It ties to the approach's target of analyzing refutation sizes and integrality gaps, avoiding relativization barriers by focusing on structural properties of random CSP instances.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 64, in run_trial
    rank = gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 29, in gaussian_elim
    factor = Fraction(1, A[i][i])
             ^^^^^^^^^^^^^^^^^^^^
   File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Debug gaussian_elimination() to handle zero pivots, perhaps by adding numerical tolerance or matrix preconditioning

---

Hardness of MCSP Under Parameterized Circuit Classes

- **Verdict**: INCONCLUSIVE
- **Bridge**: meta-complexity (MCSP, K-complexity) × hardness of approximation
- **Recorded**: 2026-04-29 02:49 UTC
- **Entry ID**: 61df9c4a8766

Statement

MCSP is hard to approximate within any constant factor for circuits of depth-3 with polynomial-size AND, OR, and MOD_p gates, implying cryptographic primitives via average-case hardness reductions.

Rationale

This conjecture builds on Hirahara's work linking MCSP to cryptographic hardness, avoiding algebraic barriers by focusing on combinatorial circuit classes. It ties to K-complexity via parameterized approximability and aligns with Impagliazzo-Oliver's techniques for hardness amplification without natural proofs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 54, in run_trial
    circuit = generate_random_circuit(n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 21, in generate_rand
    inputs = [generate_random_circuit(n // 2, depth - 1) for _ in range(2)]
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 21, in generate_rand
    inputs = [generate_random_circuit(n // 2, depth - 1) for _ in range(2)]
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 28, in generate_rand
    circuit.append([inputs[0][i], inputs[1][i]])
                    ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during circuit generation, preventing data collection. Pre-registered support condition cannot be evaluated without valid trial results. | next: Debug circuit generation recursion to handle base cases where n//2 becomes zero or one

---

Monotone CLIQUE Lower Bound via GF(2) Rank Defect

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: matroid theory × polymatroid expansion
- **Recorded**: 2026-04-29 03:21 UTC
- **Entry ID**: `550befe6a1bf`

Statement

For any monotone DNF formula F representing the k-CLIQUE_n function, the GF(2) rank defect $\Delta(F) = \log_2(m+1) - \log_2(\text{rank}_{F_2}(A_F))$ satisfies $\Delta(F) \geq \Omega(n)$, while for any DNF of size poly(n), $\Delta(F) = O(\log n)$

Rationale

The GF(2) rank defect captures the combinatorial complexity of DNF representations through matroid-theoretic lens. For k-CLIQUE, the exponential number of terms forces a large rank defect, while polynomial-size DNFs have low defect due to efficient span coverage. This measure aligns with GCT's orbit closure methods by quantifying structural rigidity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21558f1d.py", line 109, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21558f1d.py", line 76, in run_trial
    j = int(var[4:]) - 1
        ^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'x2'
```

Judge reasoning

Test crashed due to invalid literal conversion, preventing data collection to assess support fraction or counterexamples. | next: Fix the test code to handle variable parsing errors and re-run trials with multiple seeds

---

Dual Code Distance Lower Bound for Tseitin Refutations

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Proof Complexity × Coding Theory
- **Recorded**: 2026-04-29 03:54 UTC
- **Entry ID**: `8b11c41c9207`

Statement

The minimal size of a resolution refutation for Tseitin formulas on expander graphs is exponentially related to the dual distance of the linear code defined by the graph's incidence matrix.

Rationale

Tseitin formulas correspond to linear systems over GF(2), whose solution spaces form linear codes. The dual distance of these codes governs their error-correcting capabilities. By linking refutation complexity to dual code parameters, we isolate the algebraic structure of the problem from combinatorial expansion properties, avoiding natural proofs and relativization barriers. This conjecture leverages coding-theoretic techniques (e.g., dual distance bounds) alongside proof complexity methods.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 141, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 116, in run_trial
    dual_distance_value = dual_distance(A)
                          ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 67, in dual_distance
    A_inv = matrix_inverse(A)
            ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 55, in matrix_invers
    A[i][j] /= factor
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with division by zero error, preventing data collection. The crash suggests potential issues with matrix inversion but does not directly confirm or refute the conjecture. | next: Investigate why the incidence matrix is singular for tested expanders; verify graph construction parameters

---

Width-5 ABPs and NC^1 Circuit Simulation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Barrington's theorem × Algebraic branching programs
- **Recorded**: 2026-04-29 04:23 UTC
- **Entry ID**: `cd771dc62fac`

Statement

Any function computable by a depth-d NC^1 circuit can be simulated by a width-5 algebraic branching program (ABP) of size O(n^d), but there exist functions in P that require ABPs of exponential size to compute.

Rationale

Barrington's theorem establishes that width-5 ABPs can simulate NC^1 circuits, but the size of the ABP depends on the depth of the circuit. This conjecture extends the separation by proposing a polynomial-size ABP for NC^1 functions while suggesting exponential lower bounds for P functions, aligning with Barrington's techniques and avoiding barriers like relativization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 181, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 159, in run_trial
    p_function_size, p_function_result = run_p_function_simulation()
                                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 152, in run_p_functi
    abp = generate_abp_for_p_function(p_function)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 127, in generate_abp
    abp = [0] * (2**n + 2*n)
          ~~~~^~~~~~~~~~~~~~
OverflowError: cannot fit 'int' into an index-sized integer
```

Judge reasoning

Test crashed due to overflow error, preventing evaluation of conjecture | next: Optimize ABP generation algorithm to handle large n values

---

AC^0 PARITY Lower Bound via Linear Independence Invariant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: communication complexity × real algebraic geometry
- **Recorded**: 2026-04-29 05:07 UTC
- **Entry ID**: `0423a578c1a6`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the minimal number of linearly independent functions it can compute is $\Omega(\log \text{size}(C))$

Rationale

This invariant leverages the linearity of PARITY and the structural limitations of AC^0 circuits. By measuring the circuit's ability to represent linearly independent functions (a natural group action under variable permutations), we isolate the complexity inherent in computing PARITY. The logarithmic lower bound on this invariant would directly contradict the existence of small AC^0 circuits for PARITY.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1409.1534v1] Algorithms in Real Algebraic Geometry: A Survey - [2506.14060v16] Linear Geometry and Algebra - [0912.2255v3] Test ideals via algebras of p^{-e}-linear maps - [1101.3737v2] Continuous rational functions on real and p-adic varieties - [1606.00596v3] Randomized Polynomial Time Identity Testing for Noncommutative Circuits

Empirical Test

- exit code: `124`, elapsed: `240.06s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation time

---

Representation-Theoretic Rank Invariant for Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl duality, plethysm, algebraic combinatorics × Algebraic complexity theory (det vs perm separation)
- **Recorded**: 2026-04-29 05:27 UTC
- **Entry ID**: `694c0e067e6e`

Statement

There exists a representation-theoretic invariant ρ(f) of homogeneous polynomials f such that ρ(perm_n) > ρ(det_m^O(1)) for all m < n^{1.5}, and ρ is preserved under linear substitutions.

Rationale

This conjecture leverages Schur-Weyl duality and plethysm to define ρ(f) as the minimal number of symmetric tensors required to decompose f under the action of the general linear group. By bounding ρ(perm_n) via Young tableau multiplicities and contrasting it with the stable rank of det_m^O(1), it aligns with Landsberg's tensor rank approach while avoiding natural proofs barriers through algebraic geometry.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 73, in run_trial
    det_rank = representation_theoretic_rank(det_m(isqrt(n ** 1.5)))
```

```
                                                      ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 68, in det_m
    return [1] * (m - 1) + [0]
           ~~~~^~~~~~~~~
TypeError: can't multiply sequence by non-int of type 'float'
```

Judge reasoning

Test crashed due to type error in calculating m; insufficient data to evaluate conjecture | next: Fix the test code to handle integer arithmetic for m calculation

---

Submodular Matroid Rank Defect for Monotone CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: matroid theory × monotone circuit complexity
- **Recorded**: 2026-04-29 05:58 UTC
- **Entry ID**: `8c018be050c5`

Statement

There exists a submodular function μ on monotone DNF formulas such that μ(F) ≤ O(log n) for any DNF of size poly(n), μ(k-CLIQUE) = Ω(n), and μ(F ∧ G) ≤ μ(F) + μ(G) - 1 for all DNFs F, G.

Rationale

This conjecture leverages matroid rank submodularity to quantify the structural complexity of monotone DNFs. The submodularity ensures that redundant terms in the DNF do not disproportionately inflate the measure, while the CLIQUE's exponential term dependency forces μ to scale with n. This aligns with Razborov's polymatroid expansion framework but avoids algebraic geometry barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.64s`

```
erexample': 'Submodularity test failed'}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold
TRIAL: {'metric_name': 'submodularity_test', 'metric_value': None, 'instances_tested': 100, 'conject
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ed7ced5.py", line 103, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ed7ced5.py", line 103, in <genexpr>
```

```
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before collecting data; cannot verify support fraction or counterexamples | next: Fix the test to handle seed generation properly and re-run with sufficient iterations

---

SOS Integrality Gap and Secant Variety Dimension for Polynomial CSPs

- **Verdict**: INCONCLUSIVE
- **Bridge**: Algebraic Geometry × Sum-of-Squares Hierarchy
- **Recorded**: 2026-04-29 06:32 UTC
- **Entry ID**: c300a1171a3c

Statement

The integrality gap of the Sum-of-Squares (SOS) hierarchy for a polynomial constraint satisfaction problem (CSP) is bounded by the dimension of the secant variety of the variety defined by the CSP's constraints.

Rationale

Secant varieties, which capture the geometry of polynomial systems, provide a natural link to algebraic geometry. By connecting the SOS hierarchy's performance to the secant variety's dimension, we explore a novel interplay between geometric complexity and optimization. This conjecture avoids prior work on secant ranks in communication complexity and focuses on polynomial CSPs, leveraging algebraic geometry's tools to analyze SOS lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
jecture_holds': False, 'counterexample': 'SOS gap does not match secant variety dimension'}
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9c92219.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9c92219.py", line 81, in <genexpr>
    fi
```

Judge reasoning

Test crashed before producing reliable data, preventing validation of the counterexample. | next: Implement error handling for missing 'seed' field and re-run with more instances

---

Protocol-Induced Covers on Tensor-Powered Girth Gadgets Exhibit Logarithmic Multiplicity Growth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — Geometric Complexity Theory × Communication complexity class CC(f∘G^n)
- **Recorded**: 2026-04-29 07:23 UTC
- **Entry ID**: `578b4afa80c0`

Statement

For a MetricGadget G with positive asymptotic dimension (asdim_R(G) ≥ 1 at scale R = diam(G)), and its n-fold tensor power G^{⊗k} used as a gadget in the lift f∘(G$^{\{\otimes k\})}$n, every deterministic communication protocol Π computing f∘(G$^{\{\otimes k\})}$n with cost c induces a ProtocolPullback that forms a RoeCover of multiplicity m_Π ≥ log_2(k+1) for sufficiently large k, where the scale R_Π depends only on G.

Rationale

This sub-conjecture tests Axiom A3 (Asdim Amplification) in tandem with A2 (Protocol → Cover). If asdim(G^{⊗k}) grows at least linearly in k (up to constants), then A2 implies that any protocol must induce a cover whose multiplicity is at least the logarithm of the asdim-amplified structure. The ProtocolPullback must therefore reflect this coarse geometric blowup, forcing m_Π to grow with k. Since CLC is meant to bound communication complexity via coarse invariants, failure of multiplicity to grow logarithmically would invalidate the amplification mechanism needed for super-logarithmic lower bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1903.06084v8] Lifting coarse homotopies - [1803.07099v3] Varieties of coarse spaces - [2511.17227v3] A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity - [2211.07055v3] Geometric complexity theory for product-plus-power - [0608054v2] Dispersion of Mass and the Complexity of Randomized Geometric Algorithms

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 94, in run_trial
    lifted_G = lift_function(OR, G_k)
               ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 44, in lift_function
    lifted_G[i*n+j][i*n:(i+1)*n] = [f(G[i][j])]
                                    ^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 25, in OR
    return max(x)
           ^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed before producing data, preventing evaluation of the conjecture. | next: Fix the test code to handle the TypeError and re-run the experiment.

Real Rank Lower Bound for ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × ACC^0 Circuit Complexity
- **Recorded**: 2026-04-29 07:53 UTC
- **Entry ID**: `2af1f3860e05`

Statement

The real rank of the communication matrix of a Boolean function f:{0,1}^n × {0,1}^n → {0,1} is Ω(log n) if and only if f cannot be computed by a polynomial-size ACC^0 circuit.

Rationale

Williams' approach leverages algebraic rank properties of matrices to establish lower bounds. By analyzing the real rank of the communication matrix, we can derive constraints on the circuit size, extending tropical rank analysis to real semialgebraic geometry. High real rank implies structural complexity that resists ACC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.28s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation time

---

Free Entropy Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Communication Complexity
- **Recorded**: 2026-04-29 08:50 UTC
- **Entry ID**: `44428529acab`

Statement

The free entropy of the DISJOINTNESS communication matrix M_n is Ω(n), implying that its randomized communication complexity is Ω(n).

Rationale

Free entropy quantifies the complexity of noncommutative distributions. For the DISJOINTNESS matrix, its high free entropy reflects the inherent structure requiring Ω(n) communication, mirroring Forster's signed-rank technique but via free probability's noncommutative framework.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
nterexample': 'non-positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 435, 'conjecture_ho
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 435, 'conjecture_ho
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 105, 'conjecture_ho
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 105, 'conjecture_ho
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 190, 'conjecture_ho
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 0.0, 'instances_tested': 780, 'conjecture_hol
RESULT: FALSIFIED counterexample='non-positive determinant' first_failing_seed=11
```

Judge reasoning

The metric computation failed for non-positive determinants, invalidating free entropy measurements. The test's counterexamples suggest issues with metric definition. | next: Fix metric to handle non-positive determinants and revalidate computations